

Glow Up AI

Intelligent Skincare Analysis

Personalized for You

🔗 AI-POWERED

📱 FLUTTER MOBILE APP

🔗 GOOGLE GEMINI API

Team members:

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Project Advisor: Dr. Hassan Althobaiti

Why Users Need Better Skincare Guidance

- ❑ 70%+ of consumers struggle to find skincare products that suit their skin type, choose suitable products, and track their skincare progress.

1

Skin Type Confusion

- Difficulty identifying skin type
- Leads to ineffective skincare routines

2

Wrong Product Choices

- Products do not match skin needs
- Wastes money and causes poor results

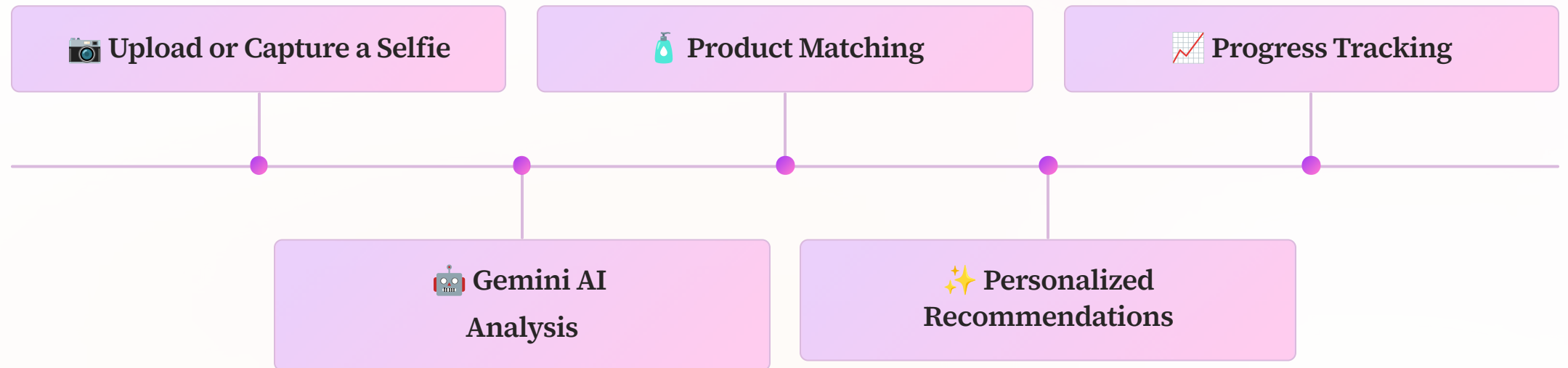
3

No Progress Tracking

- No way to monitor skin improvement
- Difficult to evaluate skincare effectiveness

Meet Glow Up AI

Analyze your skin, receive personalized recommendations, and track your skincare journey and all this is from a single selfie.



Key Features

Explore the key features that enable Glow Up AI to deliver personalized skincare analysis and recommendations.

1

AI Skin Analysis

2

Skin Type & Tone Detection

3

Personalized Daily Routine

4

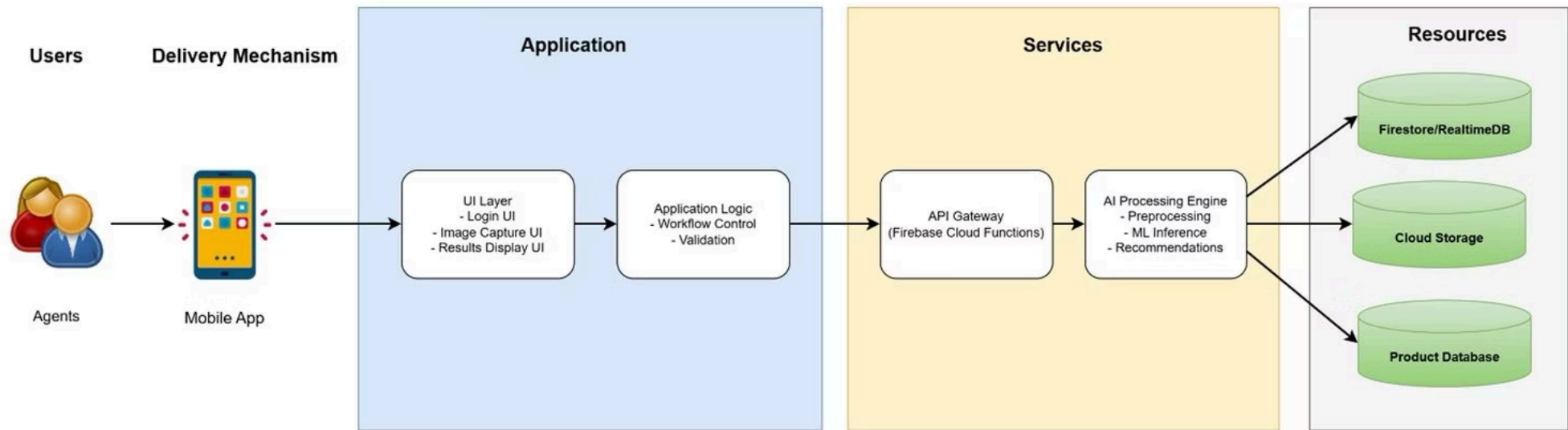
Product Recommendations

5

Progress Tracking

System Architecture

Glow Up AI is built on a modern, modular architecture that combines on-device performance with cloud-powered AI analysis. The system is designed for reliability, scalability, and a seamless user experience.



Workflow: Upload Image → AI Analysis → Product Matching → Personalized Results

Tools & Technologies

Glow Up AI uses a modern technology stack to provide AI-powered skin analysis, secure data management, and offline accessibility.



Flutter & Dart

Cross-platform mobile UI



Google Gemini API

AI skin analysis (image → data)



Firebase & Firestore

Auth & cloud database



SQLite Database

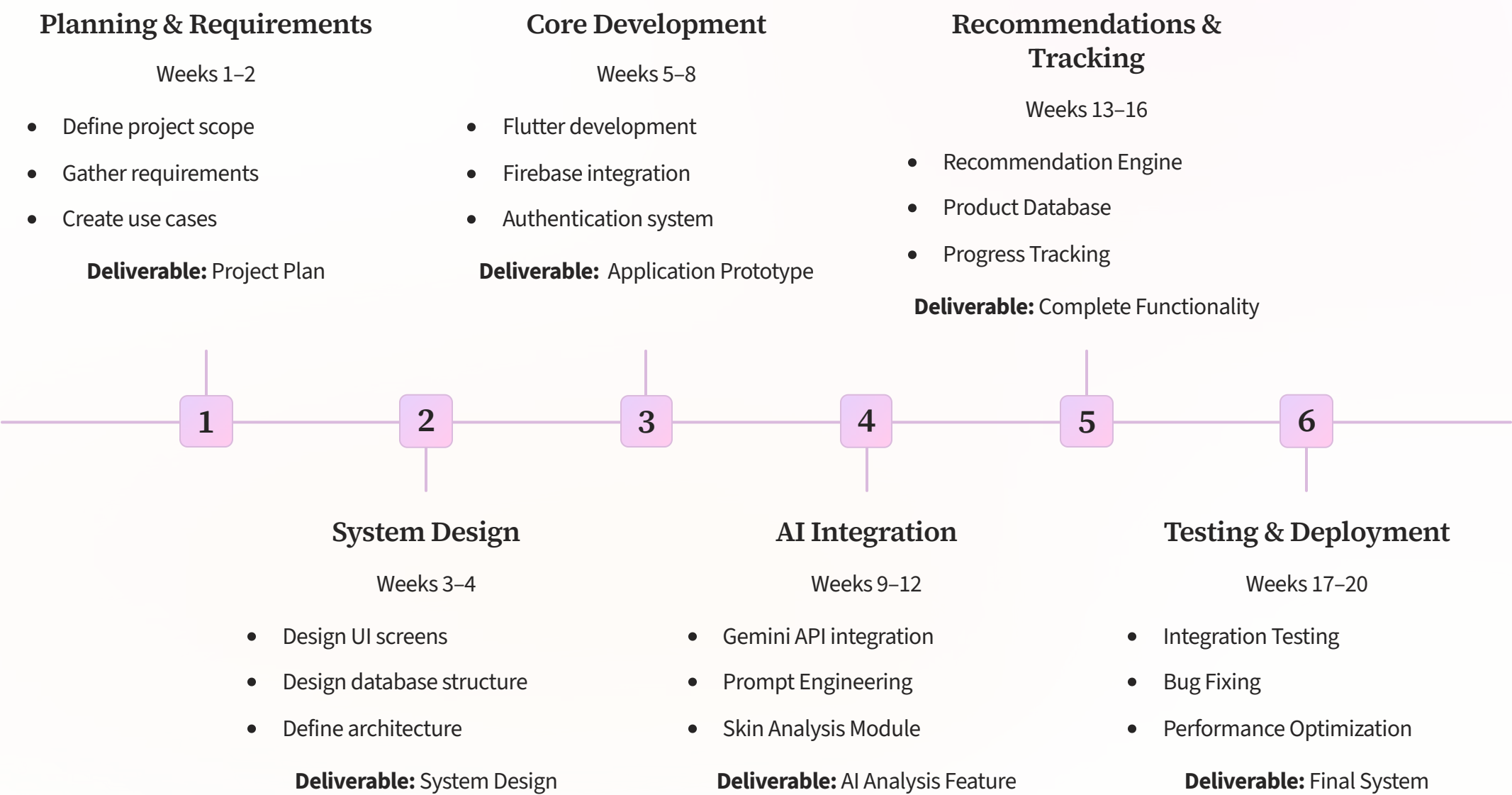
Local storage (offline support)

Why These Technologies?



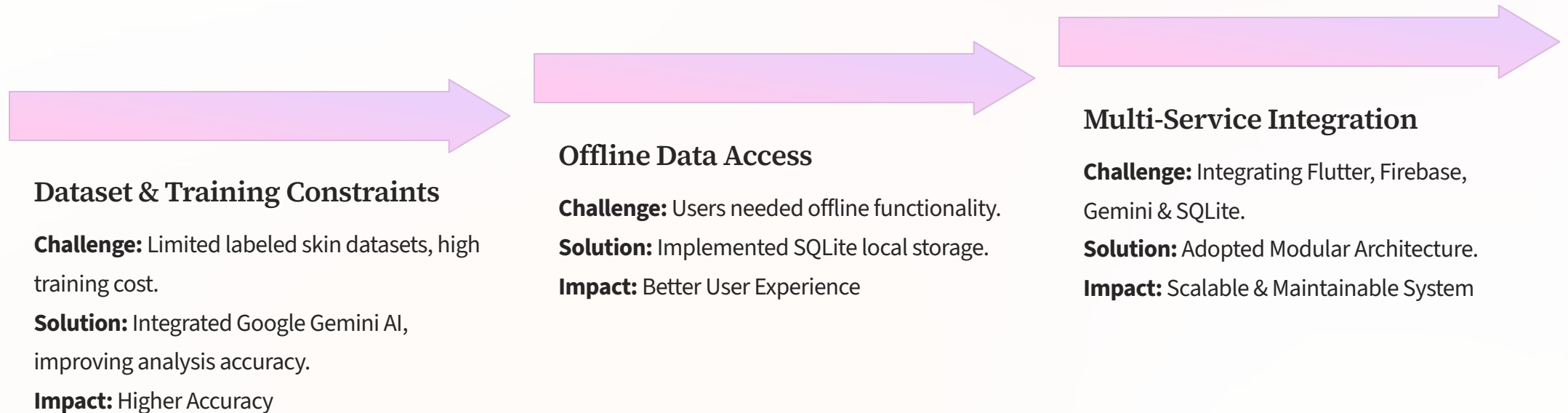
Glow Up AI Project Timeline

Project was developed over 20 weeks using an Agile approach, progressing through planning, development, AI integration, testing, and deployment phases.



Challenges & Solutions

Building an AI-powered skincare app introduced several technical challenges. The team addressed each with targeted, practical solutions that strengthened the overall architecture.



Implementation

State Management & Service Registration

```
return MultiProvider(  
  providers: [  
    ChangeNotifierProvider(create: (_) => app.AuthProvider()),  
    Provider<FirebaseFirestore>.value(value: db),  
    Provider<DatabaseHelper>.value(value: dbHelper),  
  ],  
);
```

User Authentication & Navigation Control

```
class AuthWrapper extends StatelessWidget {  
  const AuthWrapper({super.key});  
  
  @override  
  Widget build(BuildContext context) {  
    final authProvider = Provider.of<app.AuthProvider>(context);  
    if (authProvider.user != null) {  
      return const BottomNav();  
    }  
    return const WelcomeScreen();  
  }  
}
```

Firebase & Database Initialization

```
void main() async {  
  WidgetsFlutterBinding.ensureInitialized();  
  await Firebase.initializeApp();  
  
  FirebaseFirestore.instance.settings = const Settings(  
    persistenceEnabled: true,  
  );  
  
  final firestore = FirebaseFirestore.instance;  
  
  await seedProducts(firestore);  
  await seedBeautyProducts(firestore);  
  
  runApp(const MyApp());  
}  
  
Future seedProducts(FirebaseFirestore firestore) async {  
  // Seed product documents into Firestore  
}  
  
Future seedBeautyProducts(FirebaseFirestore firestore) async {  
  // Seed beauty product documents into Firestore  
}
```

Implementation

Gemini Prompt Design for Skin Assessment

```
const prompt = ""
Analyze this facial image and return ONLY valid JSON with this exact structure:
{
  "skinType": "Combination" | "Oily" | "Dry" | "Normal" | "Sensitive",
  "skinTone": "Fair" | "Medium" | "Olive" | "Brown" | "Dark",
  "undertone": "Cool" | "Warm" | "Neutral",
  "conditions": [
    {"name": "Acne", "percentage": 0-100},
    {"name": "Pigmentation", "percentage": 0-100},
    {"name": "DarkCircles", "percentage": 0-100},
    {"name": "Wrinkles", "percentage": 0-100},
    {"name": "EnlargedPores", "percentage": 0-100}
  ],
  "confidenceScore": 0.0-1.0,
  "summary": "A brief natural language summary of the analysis."
};
```

Gemini Skin Analysis Engine

```
Future<Map<String, dynamic>>
analyzeSkin(File imageFile) async {
  final url = Uri.parse(

'https://generativelanguage.googleapis.
com/v1beta/models/$_model:generate
Content?key=$apiKey',

  );

  try {
    // 1. Read and encode the image
    final imageBytes = await
imageFile.readAsBytes();
    final base64Image =
base64Encode(imageBytes);
```

Implementation

Personalized Product Matching Engine

```
final List<Product> allProducts = await
_dbHelper.readAllProducts();
print(' Total products in local database:
${allProducts.length}');
final List<Product> typeMatchedProducts =
allProducts.where((product) {
  return product.suitableSkinTypes.any(
    (type) =>
      type.toLowerCase() == analysis.skinType.toLowerCase()
    ||
      type.toLowerCase() == 'all',
  );
}).toList();
```

API Request Handling with Retry Logic

```
// 4. Send the request with Retry Logic for 503 errors
const int maxRetries = 3;
http.Response? response;

for (int attempt = 1; attempt <= maxRetries; attempt++) {
  response = await http.post(
    url,
    headers: {'Content-Type': 'application/json'},
    body: body,
  );
}
```

Implementation

User Registration & Profile Creation

```
UserCredential result = await
_auth.createUserWithEmailAndPassword(
  email: email,
  password: password,
);
if (result.user != null) {
  // Update Firebase Auth display name
  await
  result.user!.updateDisplayName(name);

  // CRITICAL FIX: Create a Firestore
  document for the new user
  await FirebaseFirestore.instance
    .collection('users')
    .doc(result.user!.uid)
    .set({
      'displayName': name,
      'email': email,
      'createdAt':
        FieldValue.serverTimestamp(),
    });
}
```

Personalized Skincare Routine Generation

```
if (analysis.conditions.any((c) => c.name.toLowerCase() == 'acne')) {
  morningSteps.add(
    RoutineStep(
      stepNumber: 2,
      title: 'Treatment Serum (Acne)',
      description:
        'Apply a salicylic acid or benzoyl peroxide serum to target active acne.',
    ),
  );
}

if (analysis.conditions.any(
  (c) => c.name.toLowerCase() == 'enlargedpores',
)) {
  morningSteps.add(
    RoutineStep(
      stepNumber: 3,
      title: 'Niacinamide Serum',
      description:
        'Apply niacinamide serum to reduce the appearance of pores and improve skin
        texture.',
    ),
  );
}
```

Testing & Evaluating

1

Functional Testing

What Was Tested: User Registration & Login, Skin Analysis, Progress Tracking, Product Ratings, Privacy Settings, Offline Mode

Result: 20/20 test cases passed successfully

2

AI Analysis Evaluation

What Was Tested: Skin Type Detection, Skin Tone Identification, Undertone Detection, Skin Condition Recognition

Result: Overall AI accuracy reached 75%

3

Performance & Reliability

What Was Tested: Application Startup Time, Login Response Time, AI Analysis Processing Time, Memory and Storage Usage

Result: Stable performance with no crashes during testing. Average AI analysis time: 4.8 seconds

4

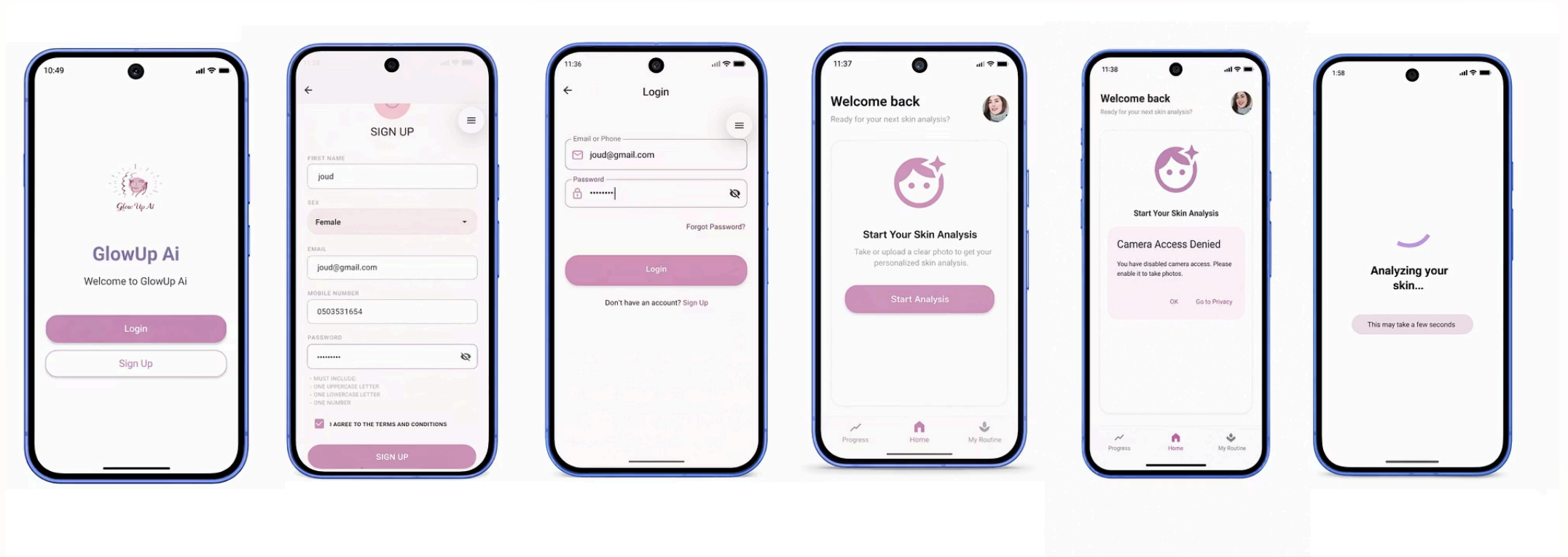
Usability Testing

What Was Tested: Ease of Navigation, Interface Design, Recommendation Usefulness, User Experience

Result: Users reported positive feedback regarding usability and interface clarity

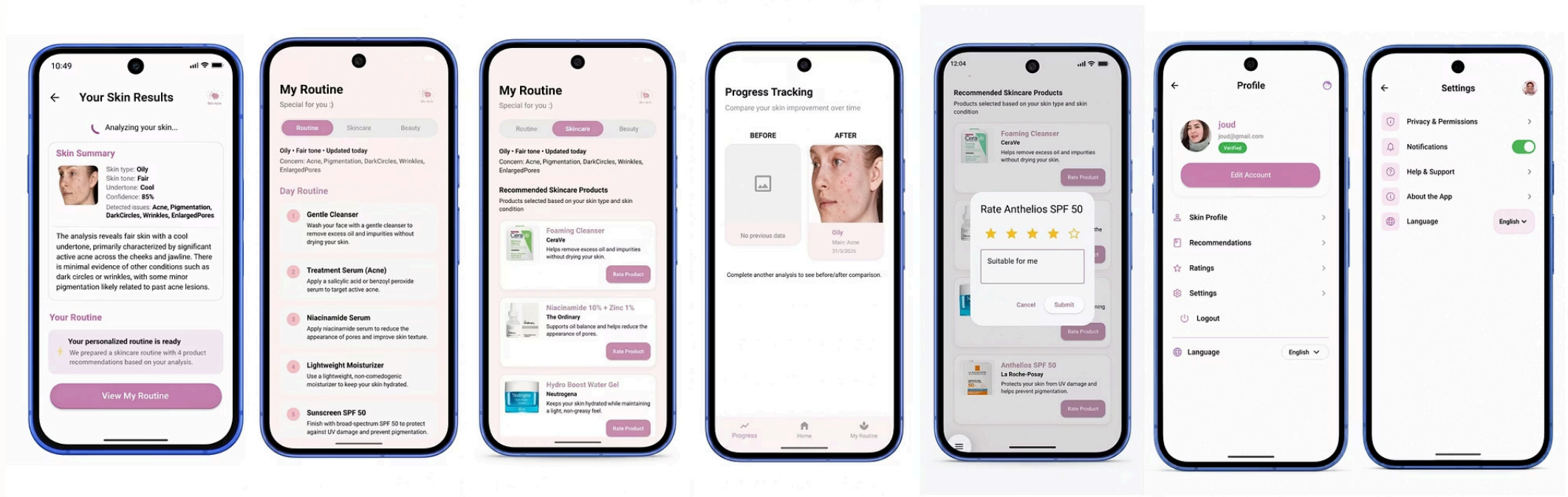
User Interfaces

All core features were thoroughly tested across registration, login, image upload, and skin analysis.



User Interfaces

All core features were thoroughly tested across routine generation, product recommendations, progress tracking, ratings & reviews, and privacy settings.



Looking Ahead: Future Enhancements

Glow Up AI is a strong foundation with a clear roadmap for growth. The following enhancements will improve the system's intelligence, personalization, and user experience.



Environmental Analysis

Use weather and environmental data to improve skincare recommendations.



AI Voice Assistant

Hands-free interaction for users to query their skin data and receive spoken guidance.



Wearable Integration

Sync with smartwatches to correlate skin health with sleep, stress, and activity data.



Expanded Product Database

Broader brand coverage and real-time product availability for more accurate recommendations.



Enhanced AI Capabilities

Deeper condition detection, ingredient analysis, and personalized regimen optimization.



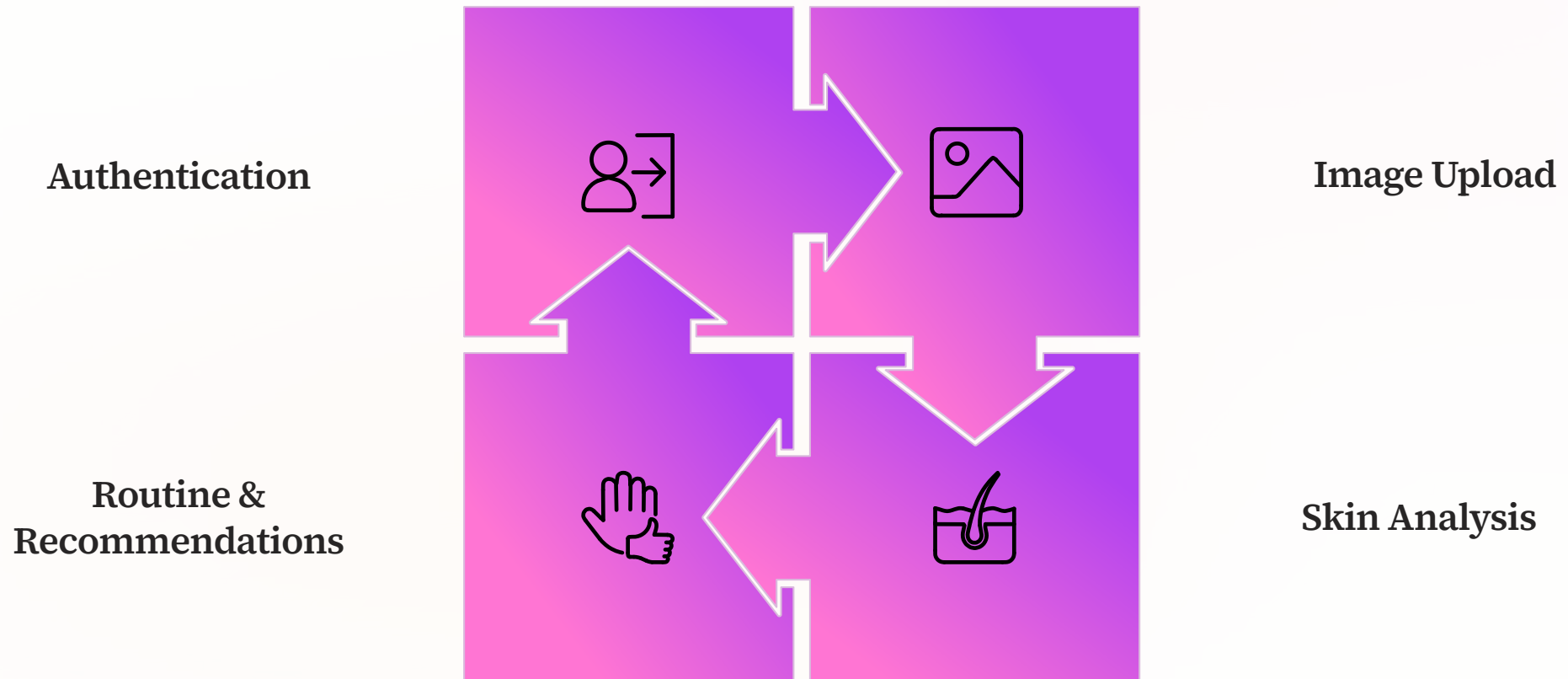
Additional Platform Support

Expand support to web and desktop platforms.

✓ These enhancements will transform Glow Up AI into a more intelligent, personalized, and comprehensive skincare platform.

Live Demo

All core features were thoroughly tested across registration, login, image upload, skin analysis, routine generation, product recommendations, progress tracking, ratings & reviews, and privacy settings. Integration testing confirmed seamless end-to-end functionality.



Conclusion & Acknowledgment

Glow Up AI combines AI and mobile technology to provide personalized skincare analysis, recommendations, and progress tracking, helping users make better skincare decisions.

Special thanks to Dr. Hassan Althobaiti for his guidance and support throughout this project.

Thank You!